

Implementing Gravitational Wave Emission in N-body Simulations: Applications to Hierarchical Triples

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Hierarchical Triple Systems

- Hierarchical Triple Systems (HTSs) consist of an inner and an outer binary.
- The stability of these systems (Gupta et al., 2020) is dependent on the condition beyond which there may be chaotic instability.

$$\frac{a_{\text{out}}}{a_{\text{in}}} > \frac{2.8}{1 - e_{\text{out}}} \left[\frac{m_{\text{out}}}{m_{\text{in}}} \frac{1 + e_{\text{out}}}{(1 - e_{\text{out}})^{\frac{1}{2}}} \right]^{\frac{2}{5}} \quad (4)$$

- From here on, we shall describe elements in the system using the subscripts 'in' and 'out' for the inner and outer binaries, respectively. Also, masses m_1 , m_2 form the inner binary and m_3 is the perturber.

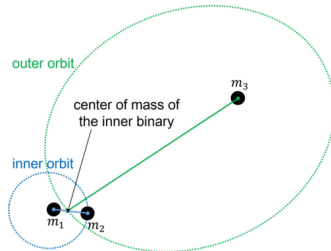


Figure 2: Simple Diagram of HTS
(Gupta et al., 2020)

Background & Related Work

For a detailed analysis of hierarchical triple dynamics and the frequency-domain signatures of eccentric binaries, see our earlier work (preprint, article under peer-review):

M. Chandrakumar, A. Kenath, R. Sharma
Dynamical Evolution of Hierarchical Triple Systems and their Gravitational Wave Detectability
Preprints.org (2026)
DOI: 10.20944/preprints202601.1591.v1



Strain from a KL active inner binary

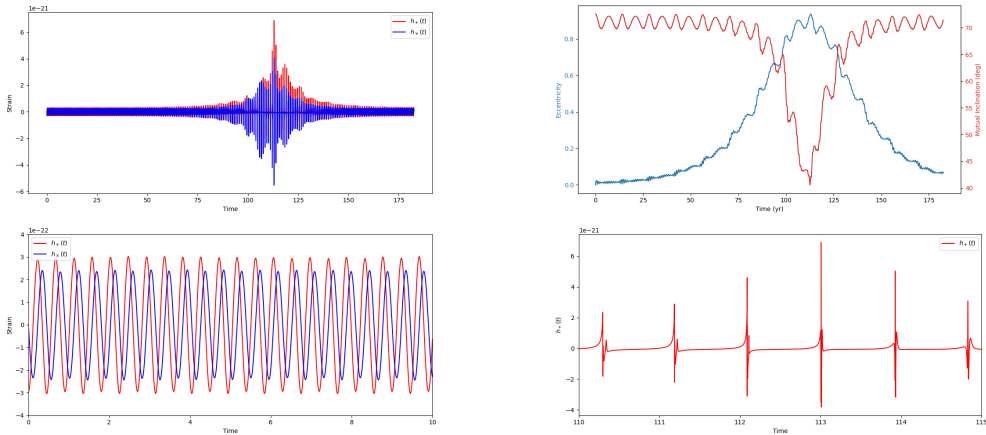


Figure 5: *Top left*: Strain, *Top right*: eccentricity-inclination cycle, *Bottom left*: Strain of binary of first 10 days and *Bottom right*: Peak strain of a hierarchical triple system $m_1 = 10M_{\odot}$, $m_2 = 10M_{\odot}$, $m_3 = 10^3M_{\odot}$, $a_{in} = 0.05AU$, $a_{out} = 1AU$ observed at a distance of $R = 8kpc$ and an observer angle of $(\chi, \iota) = (0^{\circ}, 60^{\circ})$

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